

Zimu Wang

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Education

University of California, Berkeley, Berkeley, CA

Graduation: May 2027

- Bachelor of Arts in Computer Science, GPA: 3.92/4.0
- Coursework: Neural Networks, Deep Reinforcement Learning, Computer Vision, Optimization, Probability Theory, Databases, Data Structures & Algorithms

Experience

ByteDance, LLM Engineer Intern

Summer 2026

RSIHub: github.com/simple-agent-lab/RSIHub

- Drove RSIHub's development as the lead developer, delivering an open-source **recursive self-improvement** framework that evolves agent prompts, skills, harnesses, and code.
- Architected a modular evolution engine that isolates mutable agent components from fixed evaluators and records every candidate generation in Git, preserving score integrity and commit-level reproducibility.
- Shipped a plugin operator SDK with automatic discovery, subprocess isolation, and declarative schemas, enabling new operators and strategies without core changes.
- Demonstrated best full-benchmark gains of **+13.5 pts** on Terminal-Bench 2 with MiniSWE (56.2% → 69.7%) and **+9.3 pts** on Tau3 Banking with Codex (24.7% → 34.0%) across four self-improvement strategies and all 186 tasks in the two benchmark suites.

Simple Long Horizon Agent: github.com/simple-agent-lab/simple-long-horizon-agent

- Shipped an on-demand skill runtime with package discovery, compact prompt menus, explicit skill selection, and a bundled reusable skill library.
- Unified four agent presets behind a composable AgentSession/toolset API, standardizing session and tool lifecycles while adding config-driven MCP support to evaluation runners.

Shanghai Jiao Tong University, Researcher

Summer 2025 – Spring 2026

EffiSkill

- Designed and built EffiSkill, a two-stage LLM framework for program optimization that first mines reusable Operator and Meta Skills from slow/optimized program pairs, then retrieves relevant skills and composes optimization plans to automatically optimize new programs without execution feedback.
- Built the inference and evaluation stack and benchmarked all **623 EffiBench-X tasks** in both Python and C++, with optimization success rates **3.69 to 12.52 percentage points** above the strongest baseline.

CodeJudge

- Built and scaled a pipeline to train Qwen3-8B to assess code correctness with 48K examples on **8×A100 GPUs**, using **SFT and reinforcement learning**, custom rewards, and distributed verl/vLLM rollouts.
- Raised ground-truth agreement by **15 percentage points** (54% → 69%) in held-out evaluation, outperforming **Qwen3-32B (66%)** while using only **25%** as many parameters.

University of California, Berkeley, Data 8 (Foundations of Data Science) Tutor

Fall 2024

- Led weekly tutoring sessions, a discussion section, and office hours, teaching Python-based data analysis and visualization alongside statistical methods such as hypothesis testing, regression, and classification.

Publication

- *EffiSkill: Agent Skill Based Automated Code Efficiency Optimization* [arXiv]
Zimu Wang, Yuling Shi, Mengfan Li, Zijun Liu, Jie M. Zhang, Chengcheng Wan, Xiaodong Gu.
Under review via ACL Rolling Review.

Skills

Languages: Python, C++, Java, SQL, Bash

ML/Agents: PyTorch, Hugging Face Transformers, verl, vLLM, SFT, GRPO, RLHF, MCP, agent evaluation

Infra/Tools: Docker, Weights & Biases, Git, Linux, NumPy, Pandas